

```
import java.util.HashMap;

public class INSTtable {

    HashMap<String, Integer> AD, RG, IS, CC, DL;

    public INSTtable()
    {
        AD=new HashMap<>();
        CC = new HashMap<>();
        IS = new HashMap<>();
        RG = new HashMap<>();
        DL=new HashMap<String, Integer>();
        DL.put("DC", 01);
        DL.put("DS", 02);
        IS.put("STOP",0);
        IS.put("ADD",1);
        IS.put("SUB",2);
        IS.put("MULT",3);
        IS.put("MOVER",4);
        IS.put("MOVEM",5);
        IS.put("COMP",6);
        IS.put("BC",7);
        IS.put("DIV",8);
        IS.put("READ",9);
        IS.put("PRINT",10);
        CC.put("LT",1);
        CC.put("LE",2);
        CC.put("EQ",3);
        CC.put("GT",4);
        CC.put("GE",5);
    }
}
```

```

        CC.put("ANY",6);
        AD.put("START",1);
        AD.put("END",2);
        AD.put("ORIGIN",3);
        AD.put("EQU",4);
        AD.put("LTORG",5);
        RG.put("AREG",1);
        RG.put("BREG",2);
        RG.put("CREG",3);
        RG.put("DREG",4);

    }

    public String getType(String s)
    {
        s=s.toUpperCase();
        if(AD.containsKey(s))
            return "AD";
        else if(IS.containsKey(s))
            return "IS";
        else if(CC.containsKey(s))
            return "CC";
        else if(DL.containsKey(s))
            return "DL";
        else if(RG.containsKey(s))
            return "RG";
        return "";
    }

    public int getCode(String s)
    {

```

```

        s = s.toUpperCase();
        if(AD.containsKey(s))
            return AD.get(s);
        else if(IS.containsKey(s))
            return IS.get(s);
        else if(CC.containsKey(s))
            return CC.get(s);
        else if(DL.containsKey(s))
            return DL.get(s);
        else if(RG.containsKey(s))
            return RG.get(s);
        return -1;
    }

}

```

```

public class TableRow {
    String symbol;
    int address,index;
    public String getSymbol() {
        return symbol;
    }
    public TableRow(String symbol, int address) {
        super();
        this.symbol = symbol;
        this.address = address;
        index=0;
    }
    public void setSymbol(String symbol) {
        this.symbol = symbol;
    }
}

```

```
}  
  
public TableRow(String symbol, int address, int index) {  
    super();  
    this.symbol = symbol;  
    this.address = address;  
    this.index = index;  
}  
  
public int getAddress() {  
    return address;  
}  
  
public void setAddress(int address) {  
    this.address = address;  
}  
  
public int getIndex() {  
    return index;  
}  
  
public void setIndex(int index) {  
    this.index = index;  
}  
}
```